

Project Design Phase

Proposed Solution Template

Date	2 July 2026
Team ID	
Project Name	Nutrition Assistant Using MERN stack
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Many individuals struggle to maintain a healthy lifestyle due to a lack of personalized nutrition guidance, difficulty tracking daily calorie and nutrient intake, and limited access to an easy-to-use platform for meal planning and health monitoring.
2.	Idea / Solution description	A web-based Nutrition Assistant developed using the MERN Stack (MongoDB, Express.js, React.js, Node.js) that enables users to calculate BMI, track daily nutrition, monitor calorie intake, receive AI-powered meal recommendations, and visualize their health progress through an interactive dashboard.
3.	Novelty / Uniqueness	Integrates BMI calculation, calorie tracking, nutrition analysis, AI-based personalized meal recommendations, and health progress monitoring into a single responsive web application with secure authentication and an intuitive user interface.

4.	Social Impact / Customer Satisfaction	Encourages healthier eating habits, improves nutritional awareness, supports disease prevention through better diet management, and helps users achieve their fitness and wellness goals with personalized guidance and continuous progress tracking.
5.	Business Model (Revenue Model)	Basic features are available free of charge. Future revenue opportunities include premium subscription plans, personalized AI nutrition coaching, dietician consultations, advanced analytics, health reports, and partnerships with fitness centers, healthcare providers, and nutrition brands.
6.	Scalability of the Solution	The MERN Stack architecture is highly scalable. React.js provides a responsive user interface, Node.js and Express.js efficiently handle concurrent user requests, while MongoDB supports flexible and scalable data storage. The application can be deployed on cloud platforms such as AWS, Azure, Render, or Vercel to support a growing user base.